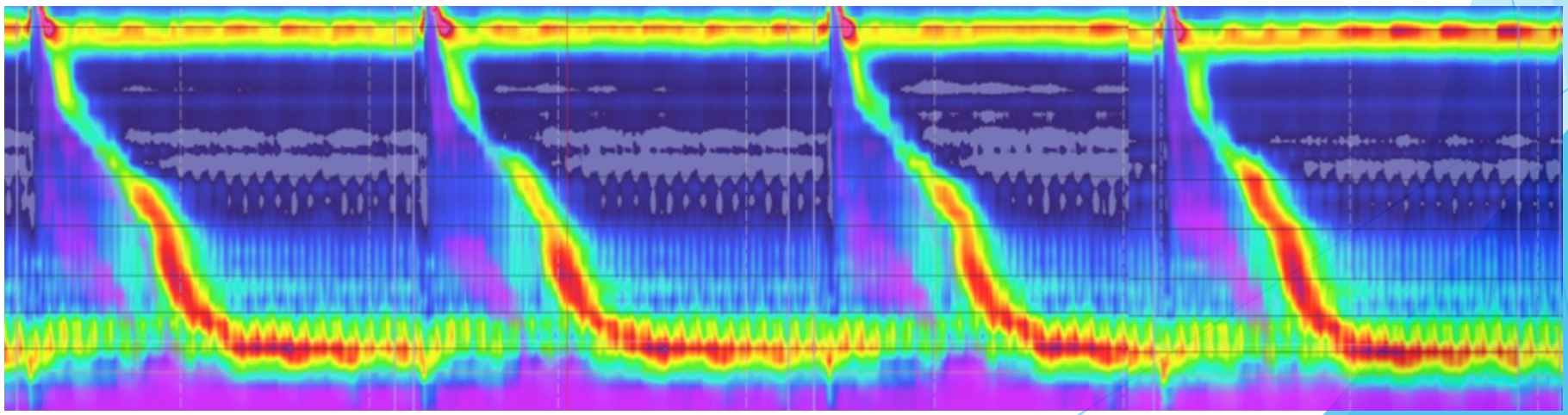
GI Motility 2

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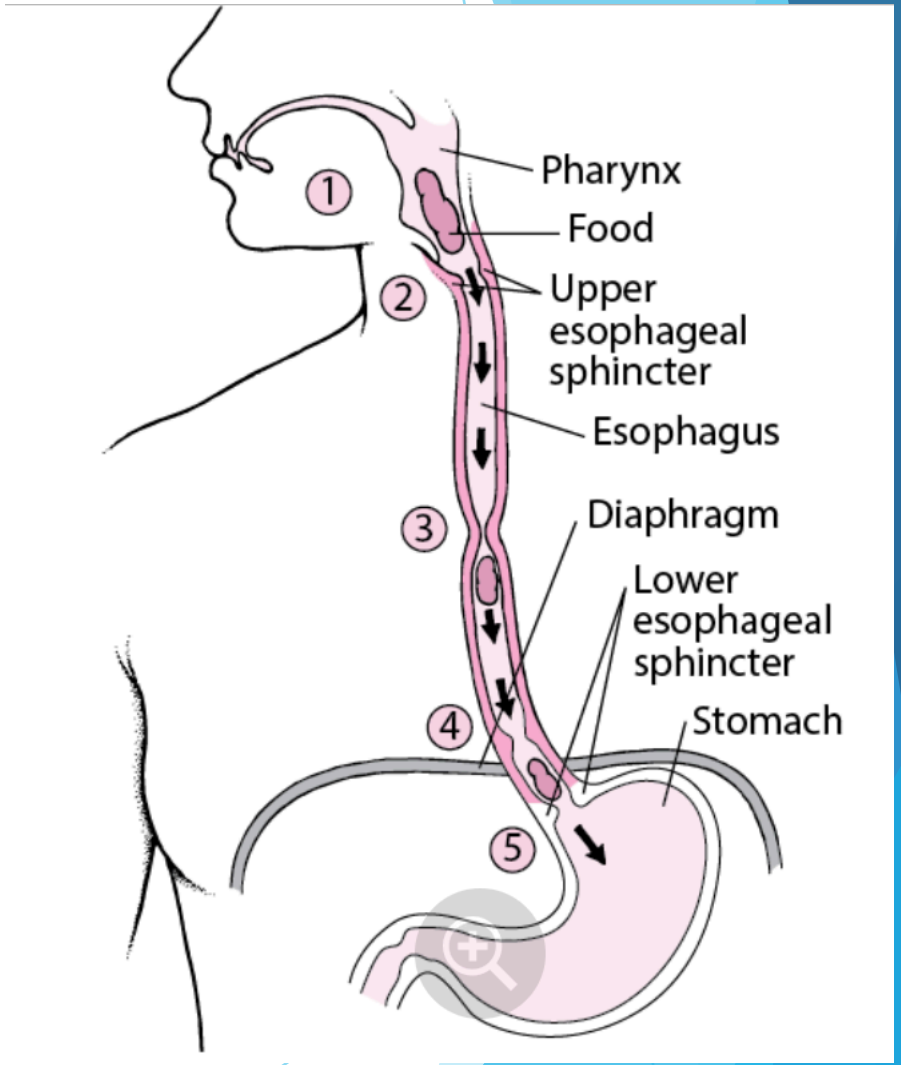
Contents

- Cellular basis and control of GI motility
- Esophageal motility
- Stomach motility
- Small intestinal motility
- Large intestinal motility
- Gallbladder and biliary tract motility

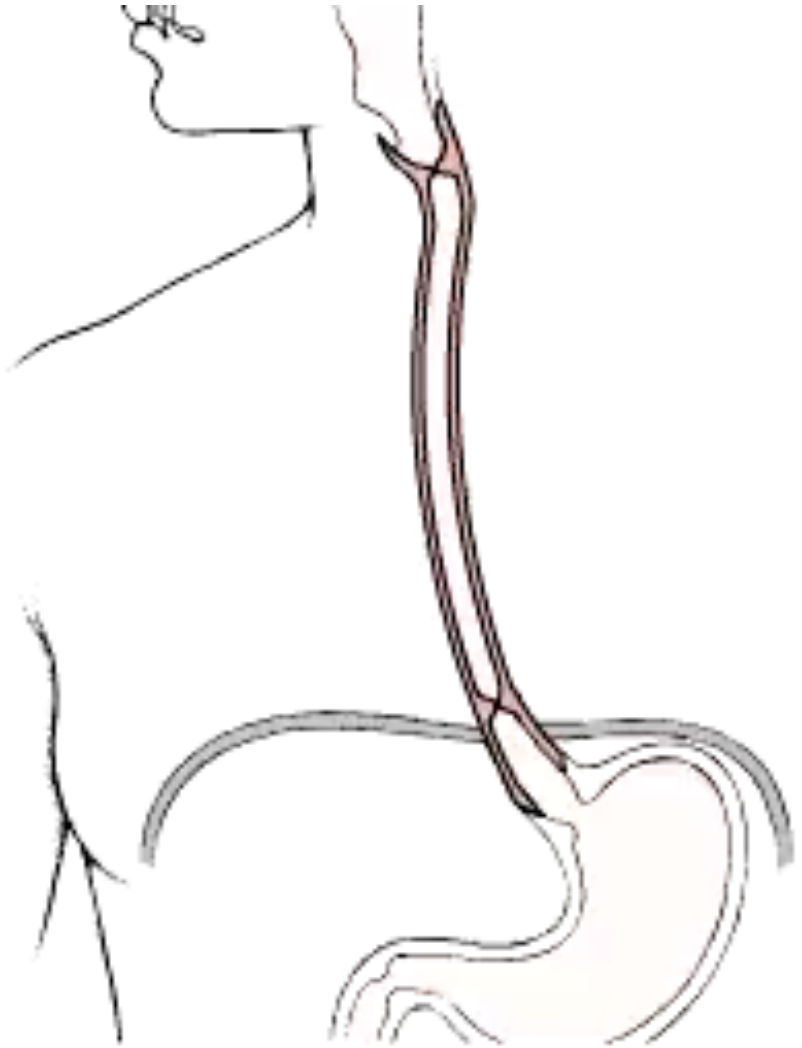
Esophageal motility

Esophageal motility

- The esophagus propels the swallowed food into the stomach.
- Upper esophageal sphincters : prevent air from entering the upper esophagus, prevent food regurgitation
- Lower esophageal sphincters : prevent gastric acid from entering the lower esophagus, prevent food regurgitation

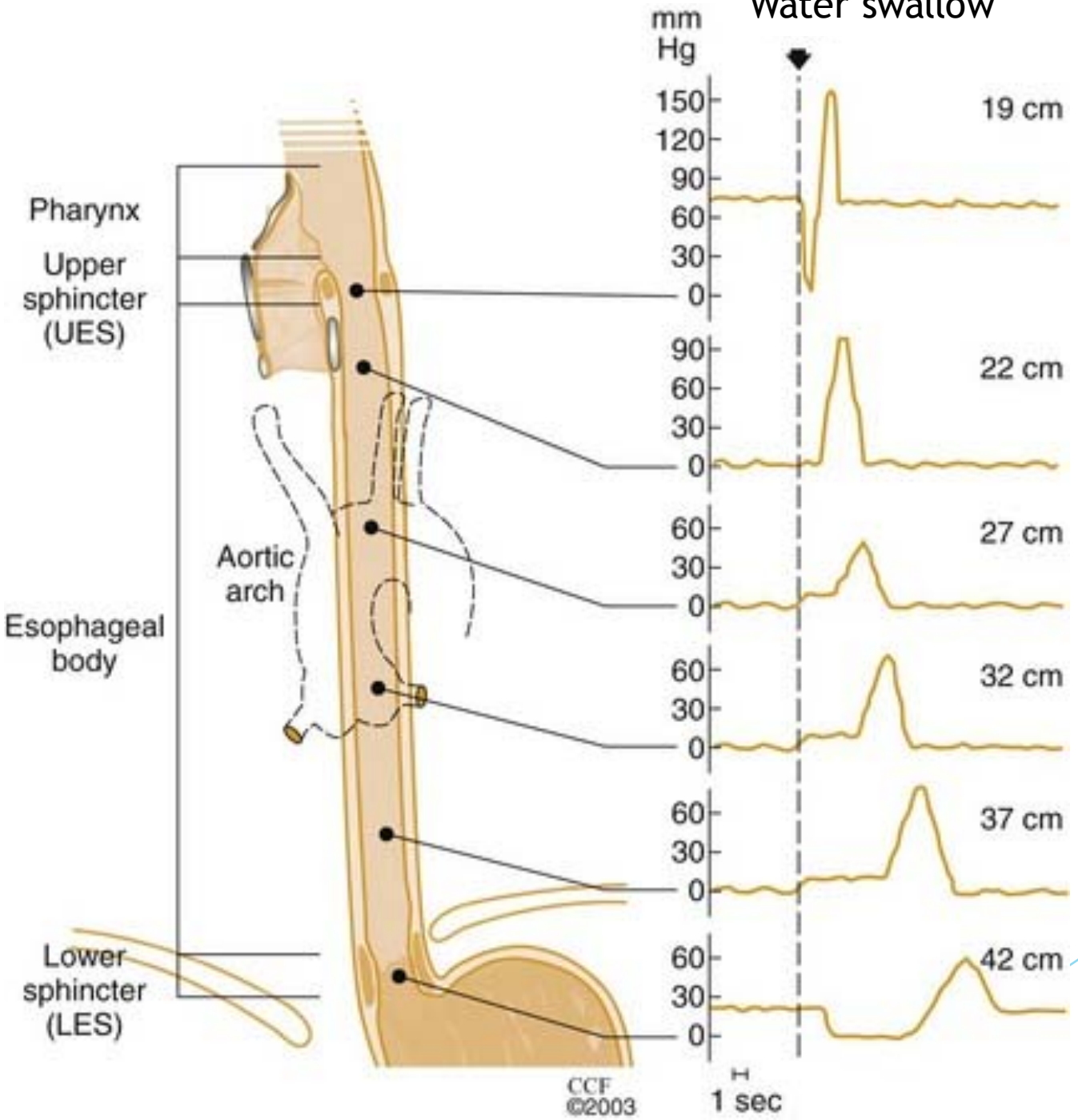


MSD MANUAL Consumer Version

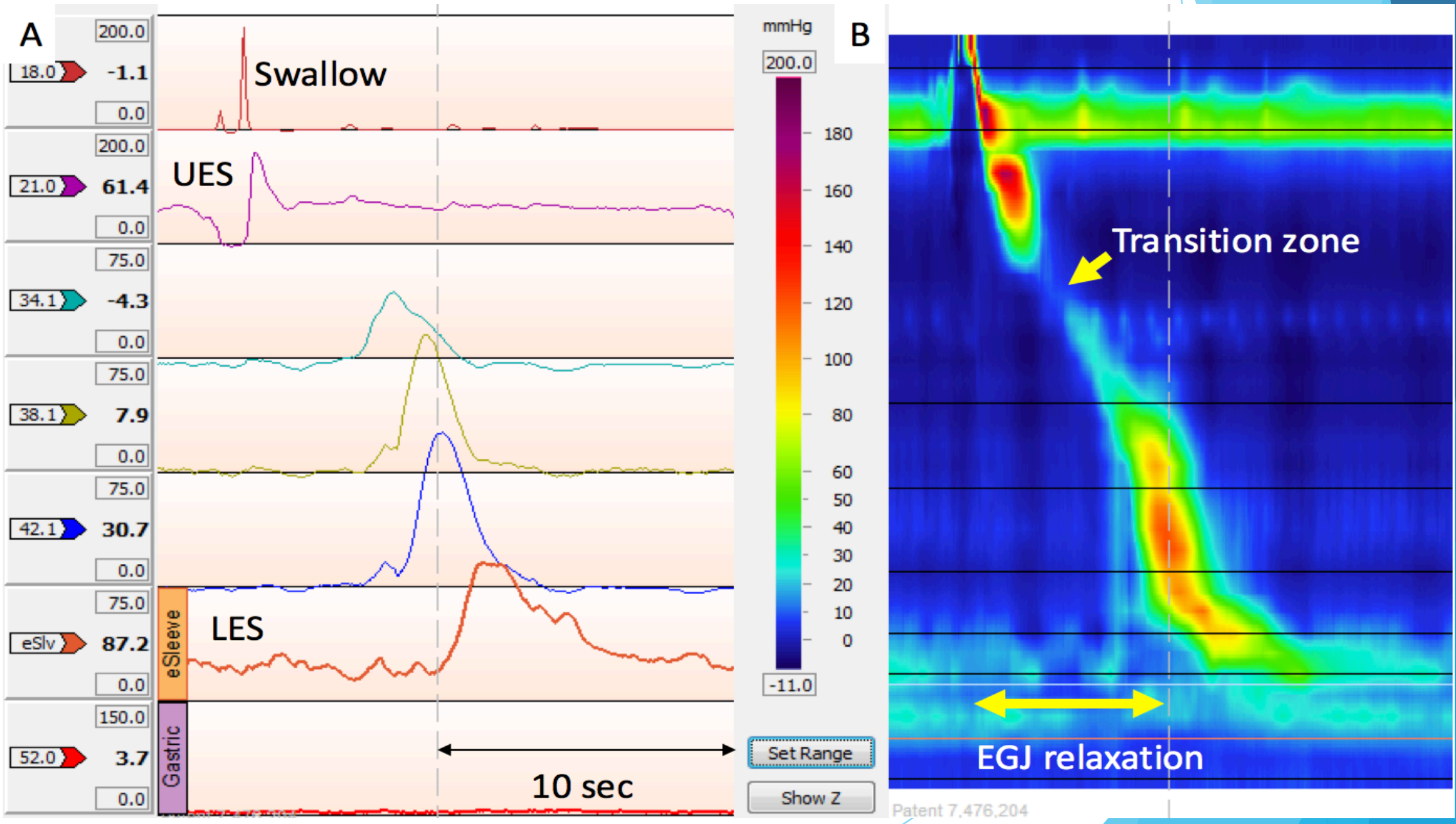


MSD MANUAL
Consumer Version

Water swallow



Esophageal manometry

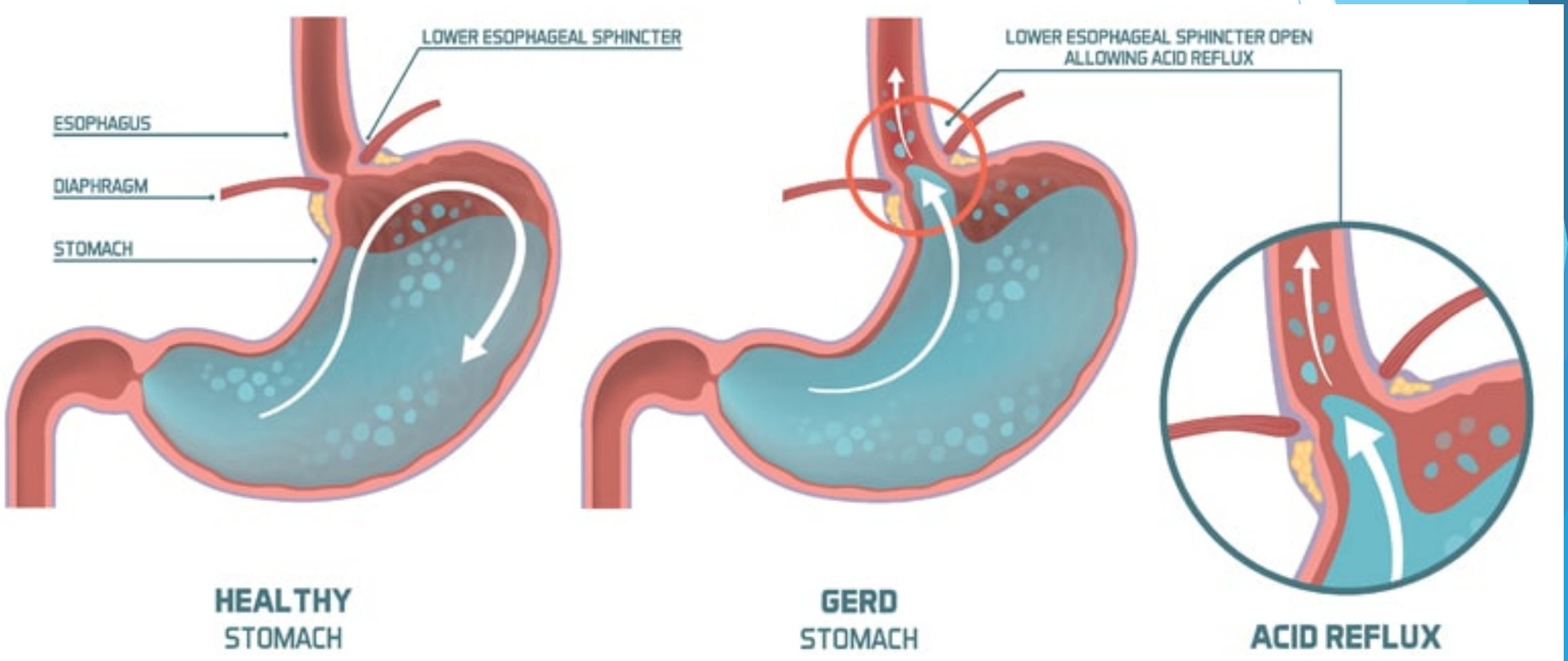


Esophageal motility : Peristalsis

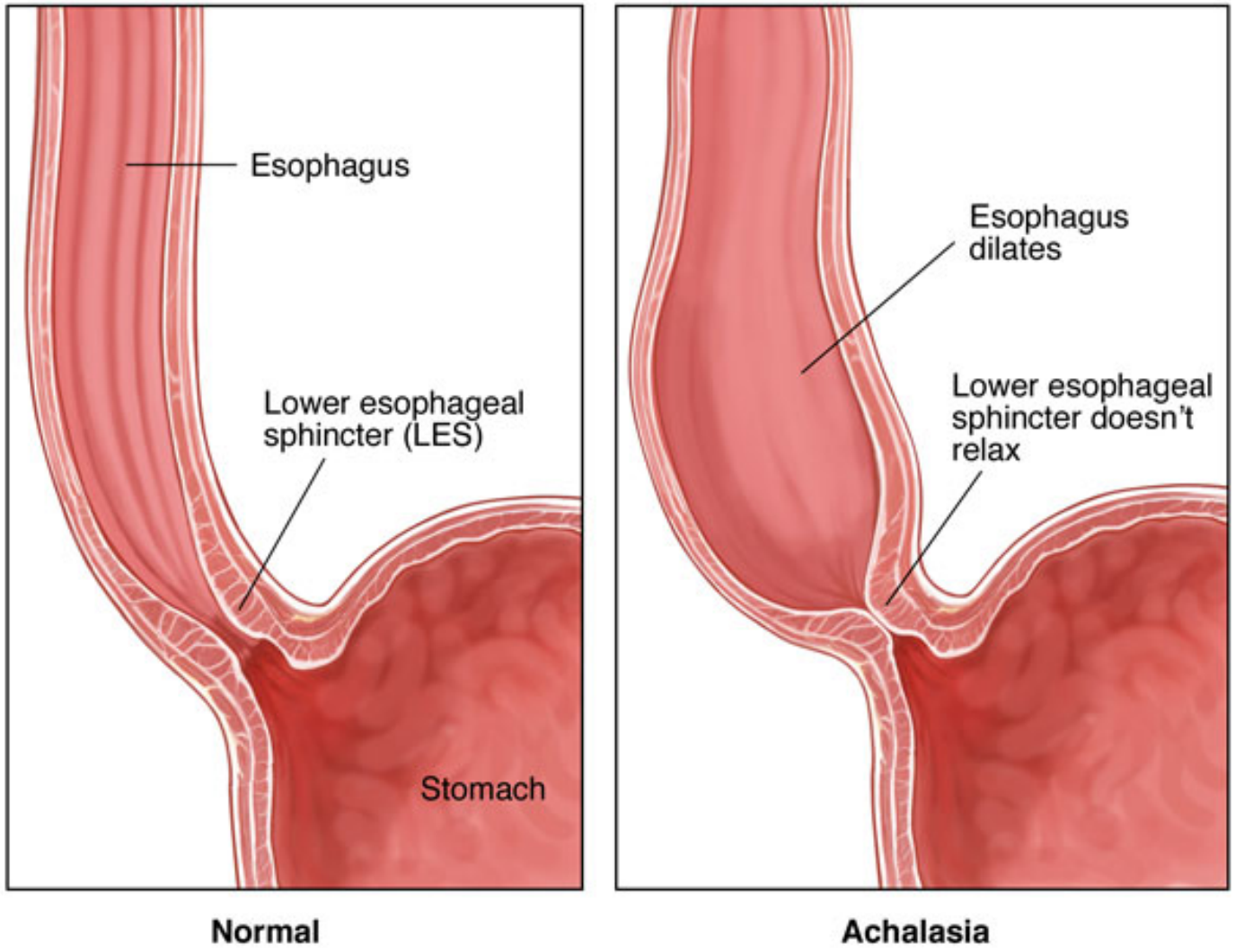
1. Primary peristalsis
   - Creates an area of high pressure behind the food bolus
   - Moves down the esophagus and propels the food bolus along
   - Gravity accelerates the movement.

2. Secondary peristalsis
   - Clears the esophagus of any remaining food

Clinical application: Gastroesophageal reflux disease (GERD)



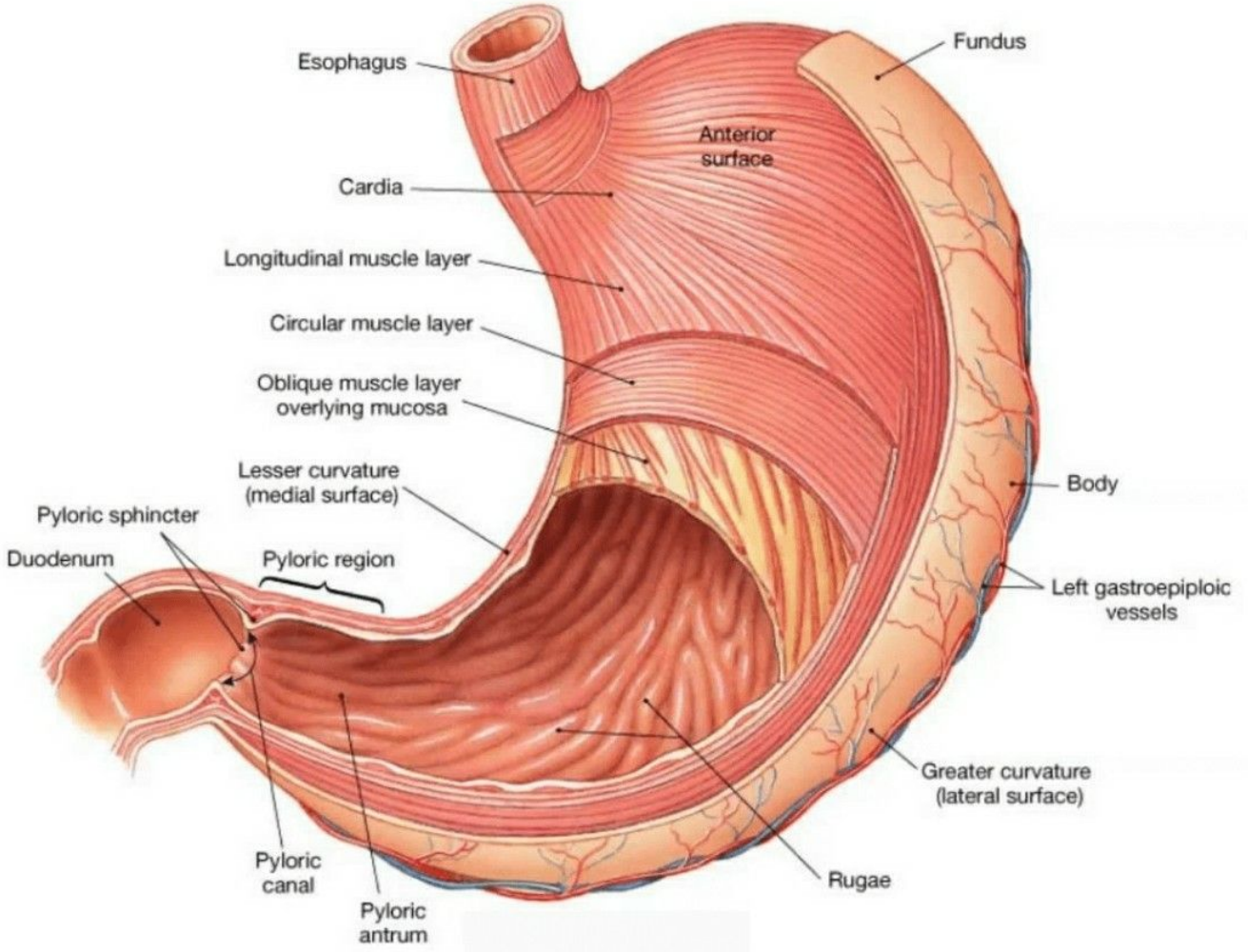
Clinical application: Achalasia

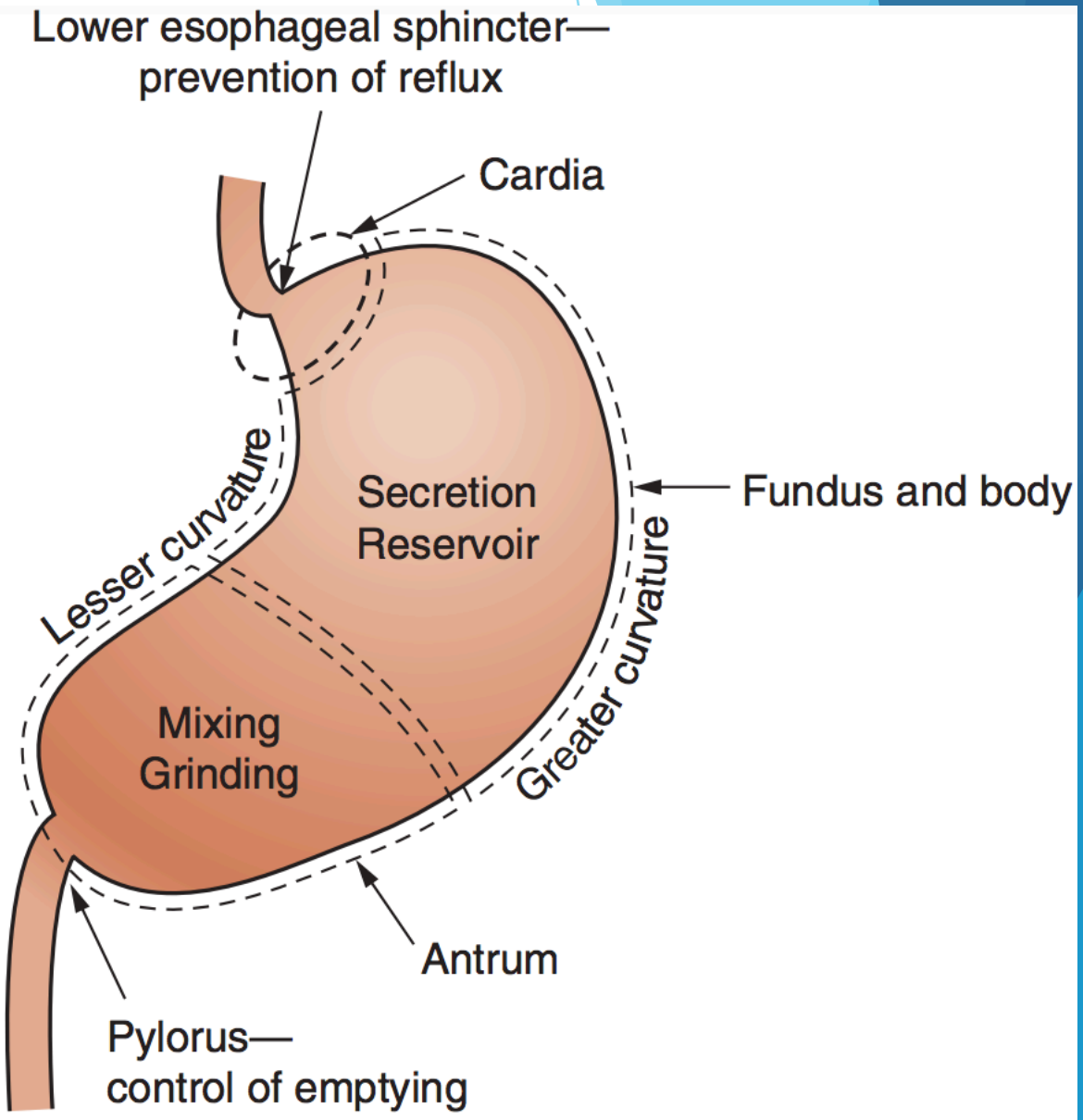


- LES failed to relax
- No peristalsis

Stomach motility

Anatomy of the stomach





Gastrointestinal Physiology. Lange Medical Books/McGraw-Hill, 2006.

Receptive relaxation

- The orad region of the stomach relaxes to accommodate the ingested meal. (Reservoir function)
- Vagovagal reflex that is initiated by distention of the stomach
- CCK participates increase the distensibility of the orad stomach.
- Gastric restriction in a surgical treatment for obesity (loss the reservoir function)

Mixing and digestion

- Occurred in the caudad region > contracts to mix the food with gastric secretions and reduces the food particles size.
- A wave of contraction closes the distal antrum. Thus, as the caudad stomach contracts, food is propelled back into the stomach to be mixed (retropulsion).
- Gastric contractions are increased by vagal stimulation and decreased by sympathetic stimulation.
- Chyme = ingested food that is mixed with the stomach secretions

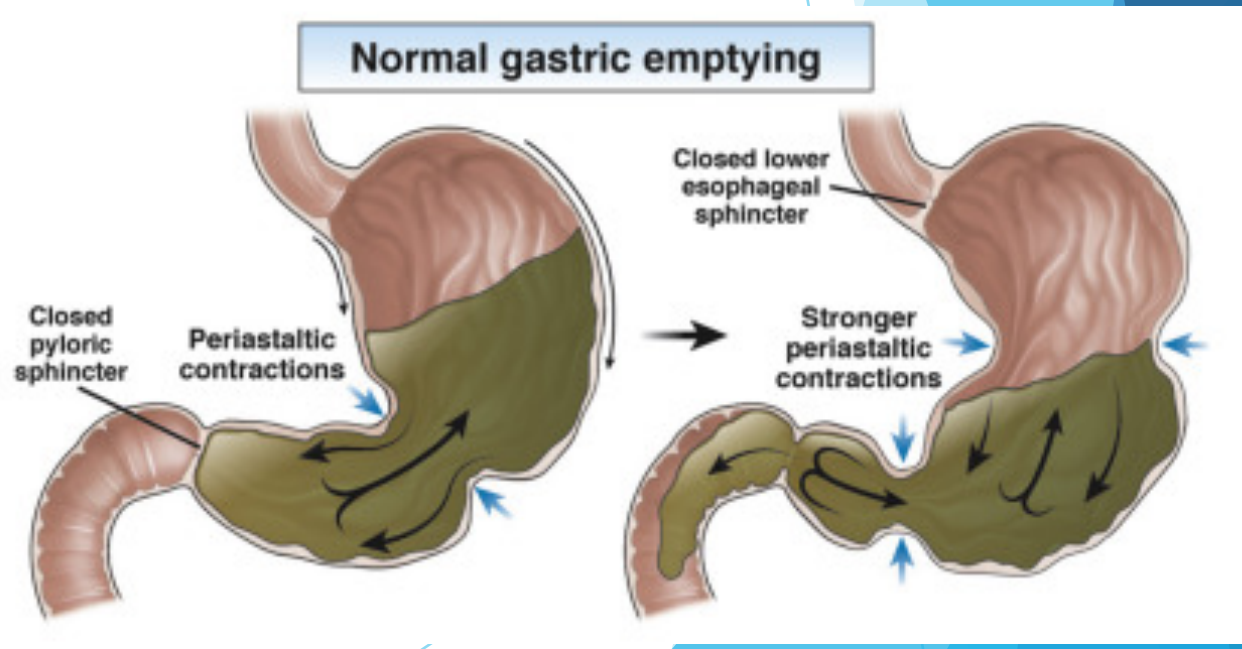
Hunger Contractions

- Occurs when the stomach has been empty for several hours or more.
- Extremely strong, often fuse to cause a continuing tetanic contraction that sometimes lasts for 2 to 3 minutes.
- Greatly increased by the person’s having lower than normal levels of blood sugar

Gastric emptying

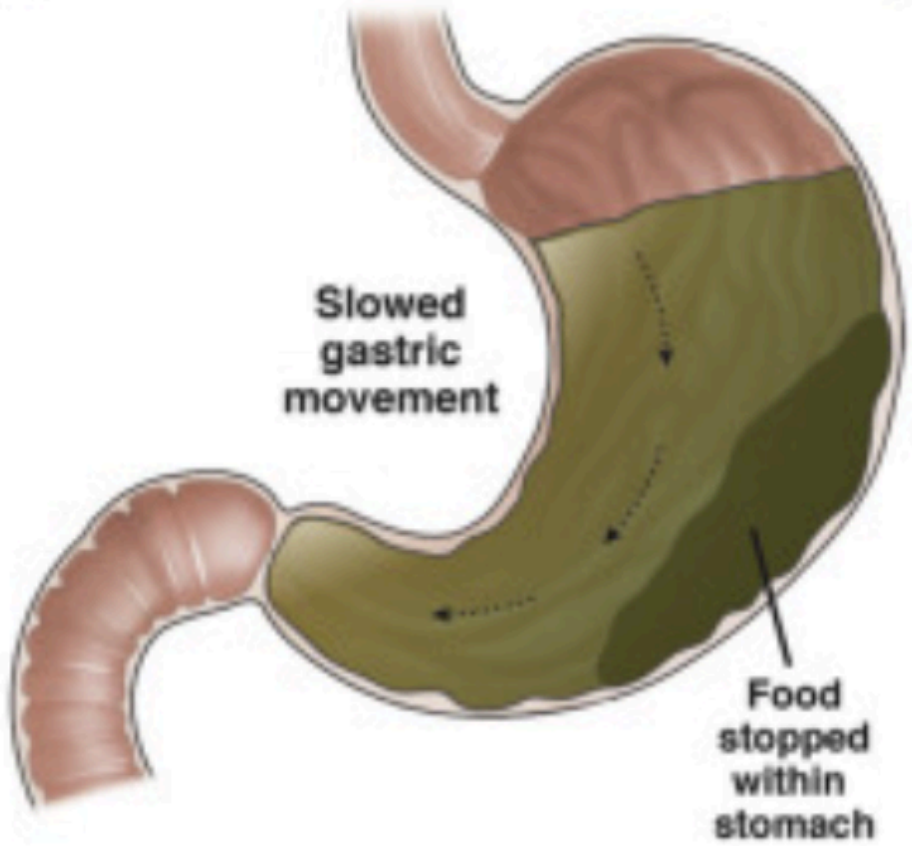
- The caudad region of the stomach contracts to propel food into the duodenum.

- "Pyloric pump": strong antral peristalsis contraction to force chyme into the duodenum.
- The rate of gastric emptying is controlled by many factors.
  - Inhibit gastric emptying: Fat, Carbohydrate, CCK, acid in duodenum
  - Promote gastric emptying: Stretching of stomach wall, gastrin



Clinical Gastroenterology and Hepatology, 17(10), 2019

Clinical application: Gastroparesis



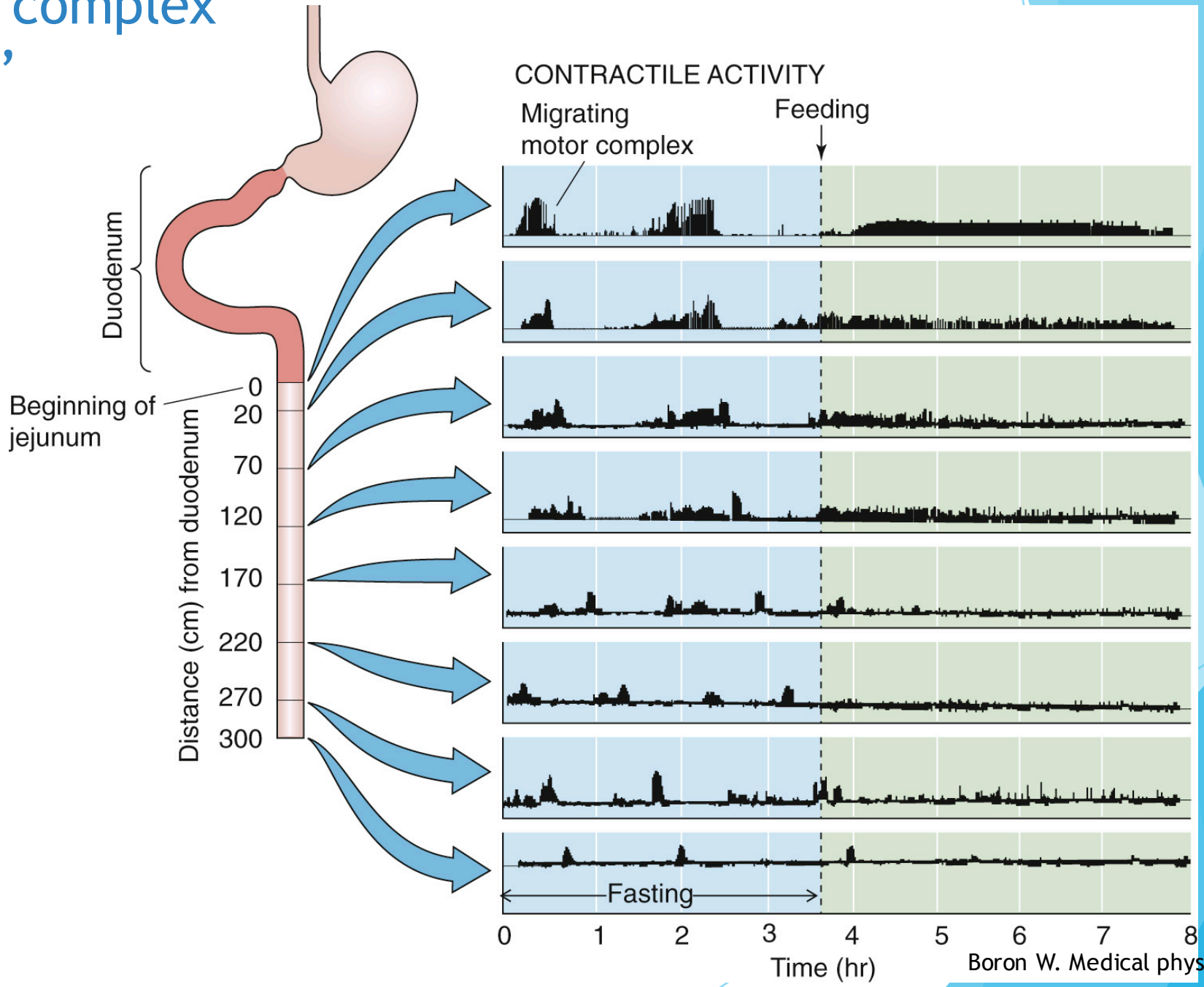
- Post surgery (Vagotomy)
- Diabetes
- Neurologic disease

Clinical Gastroenterology and Hepatology, 17(10), 2019

Migrating motor complex "MMC"

- Occur at 90-minute intervals, during fasting (interdigestive phase)
- Begin in antrum or duodenum to terminal ileum
- Mediated by motilin (released cyclically, inhibited after ingestion of food)
- "Housekeeper" - Clear the stomach of residual food in stomach and intestine

Migrating motor complex "MMC"



Boron W. Medical physiology, Elsevier, 2017

Small intestinal motility

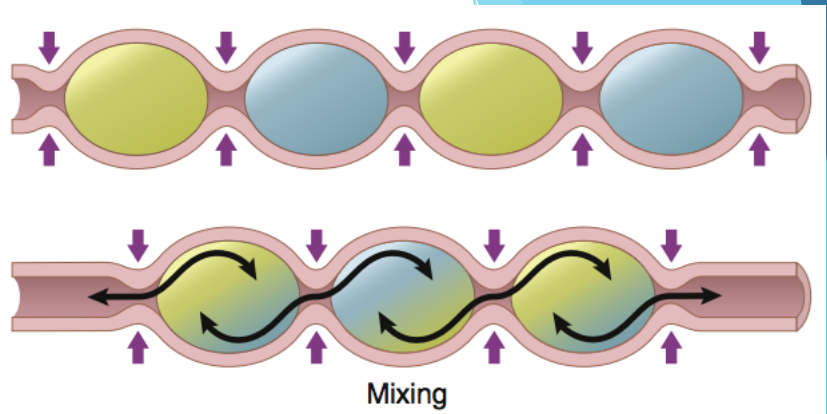
Small intestinal motility

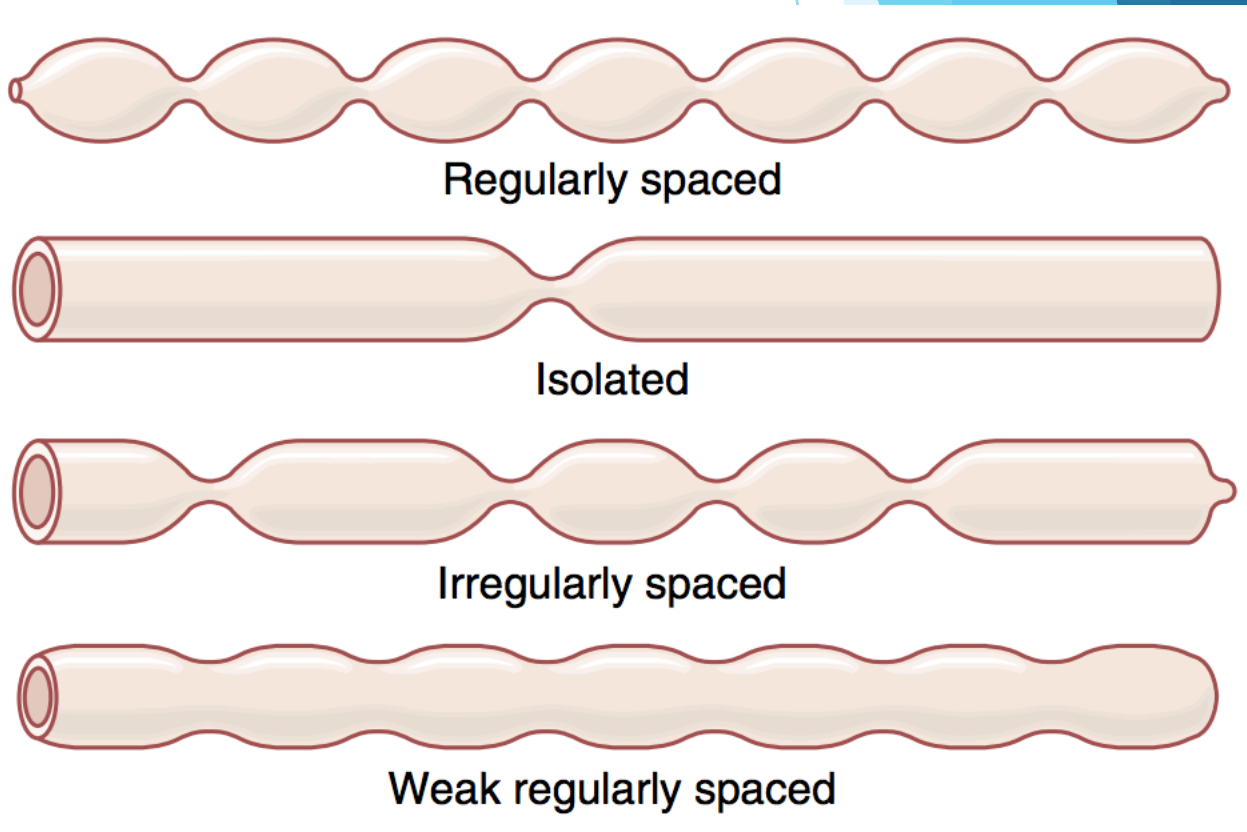
- Main functions of the small intestine: digestion and absorption of nutrients

- The small intestine mixes nutrients with digestive enzymes, exposes the digested nutrients to the absorptive mucosa, and then propels any nonabsorbed material to the large intestine.

Segmentation contractions

- Mix the intestinal contents
- A section of small intestine contracts, sending the intestinal contents (chyme) in both orad and caudad directions.
- That section of small intestine then relaxes, and the contents move back into the segment.
- This back-and-forth movement produced by segmentation contractions causes mixing without any net forward movement of the chyme.

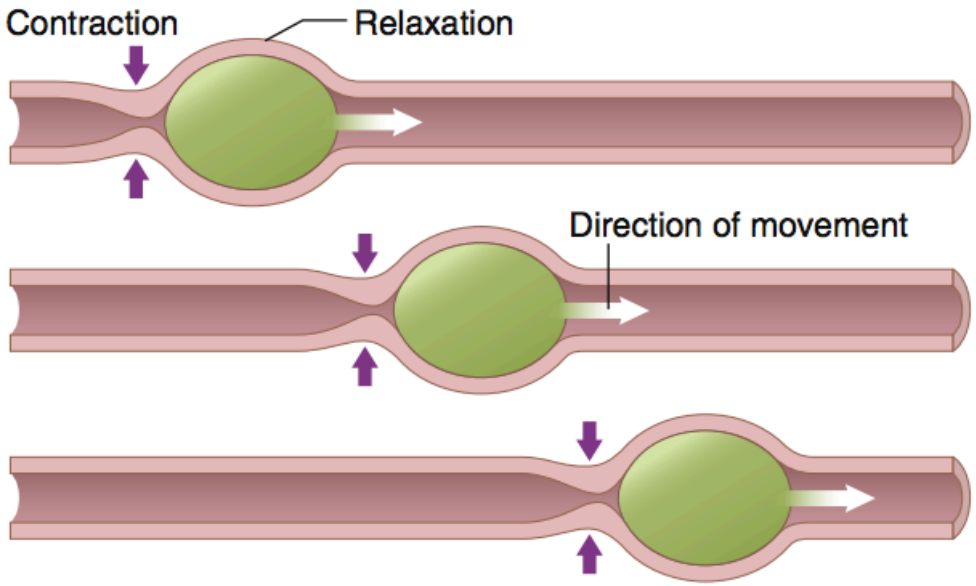




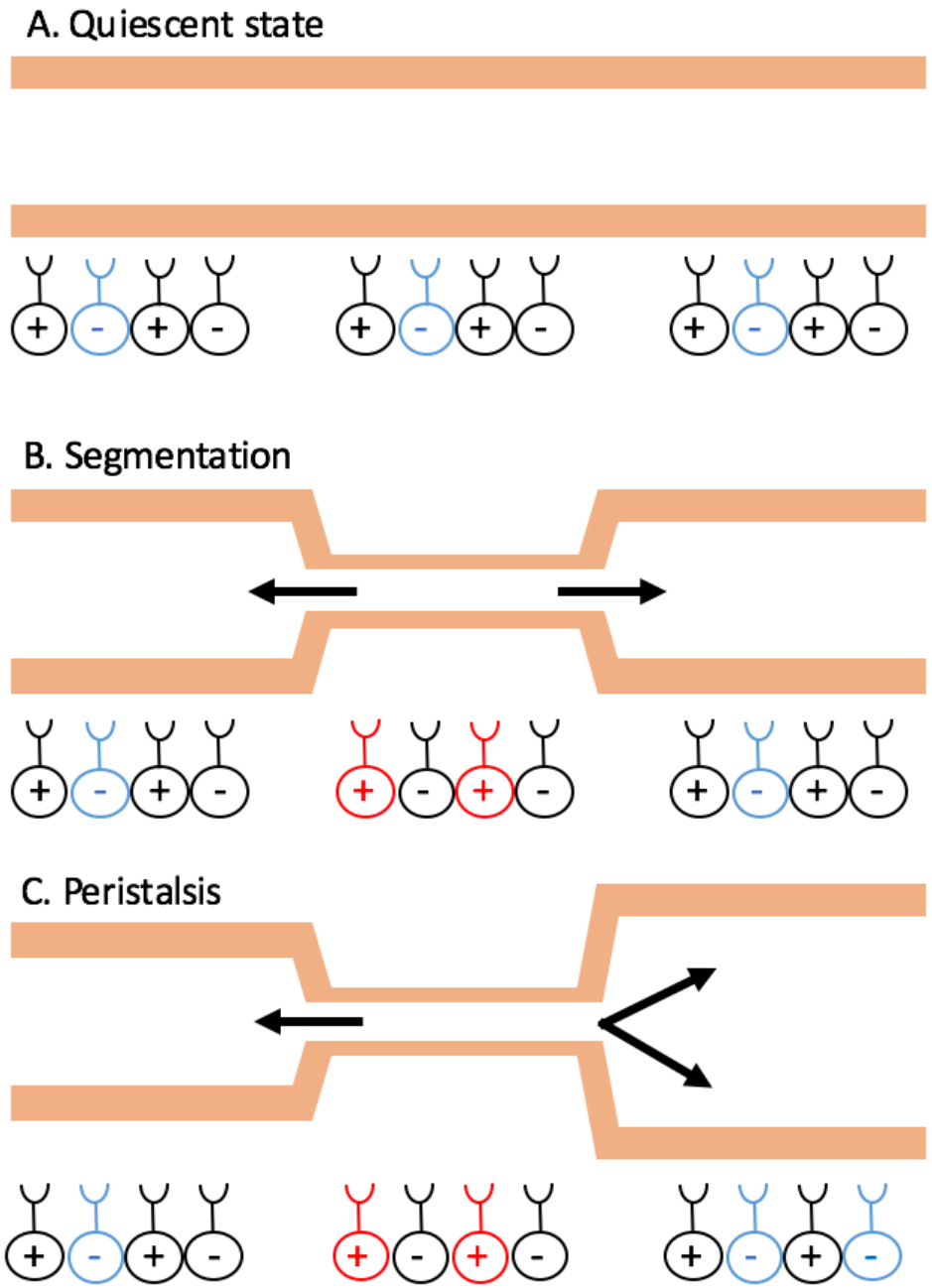
Ganong's Review of Medical Physiology, 23rd ed. McGraw-Hill Medical, 2009.
Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

Peristaltic contractions

- Propel the chyme through the small intestine toward the large intestine.
- Contraction behind the bolus and, simultaneously, relaxation in front of the bolus cause the chyme to be propelled caudally.
- The peristaltic reflex is coordinated by the enteric nervous system.



Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.



(+) Excitatory neurons: ACh, Substance P
(-) Inhibitory neurons: VIP, NO

Black: Inactive inhibitory (-) or excitatory (+) neurons
Blue: active inhibitory (-) neurons
Red: active excitatory (+) neurons

Physiology and Pathophysiology of Digestion. Morgan & Claypool; 2018.

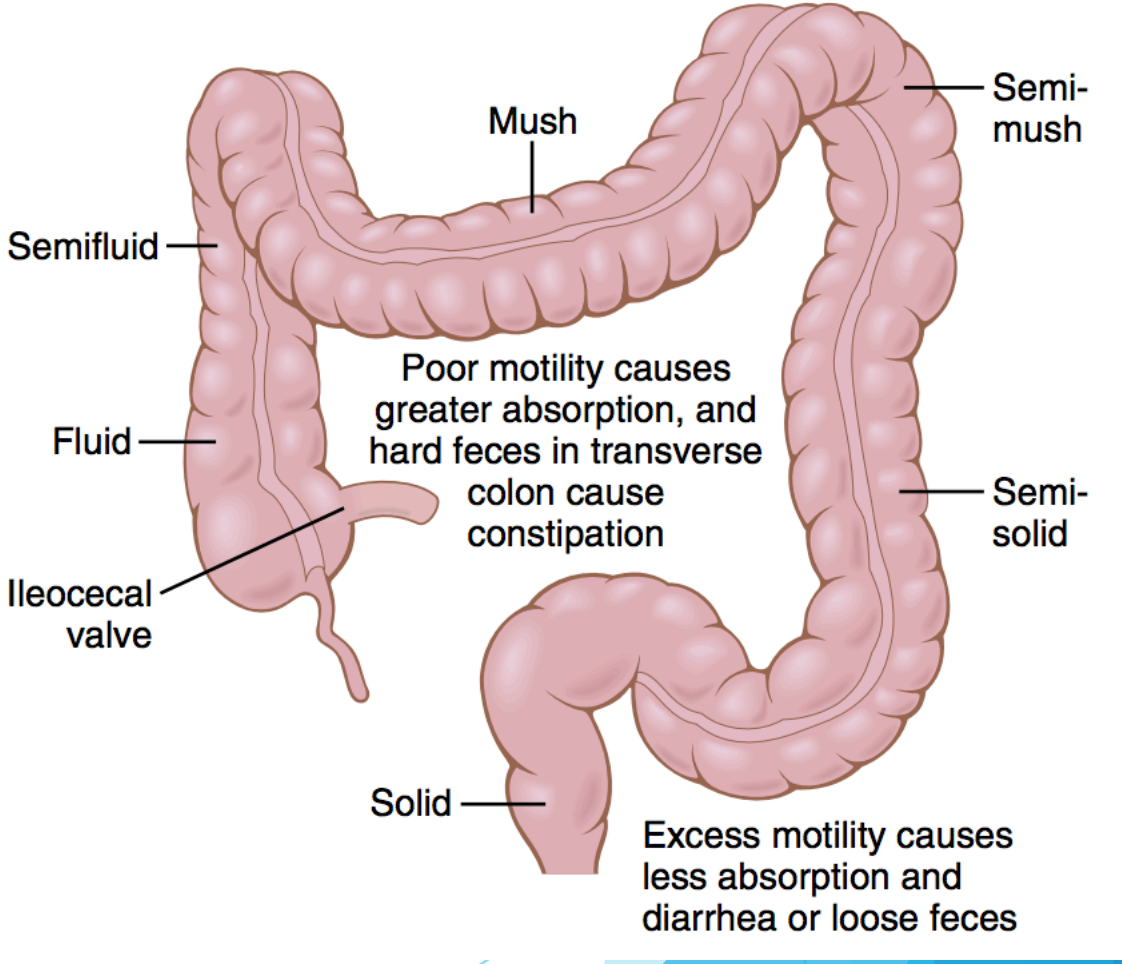
Large intestinal motility

The colon

- Functions of the colon
  - Absorption of water and electrolytes from the chyme to form solid feces
  - Storage of fecal matter until it can be expelled

- Haustra, or saclike segments, appear after contractions of the large intestine.

- Mixing Movements > Haustrations
- Propulsive Movements > Mass Movements



Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

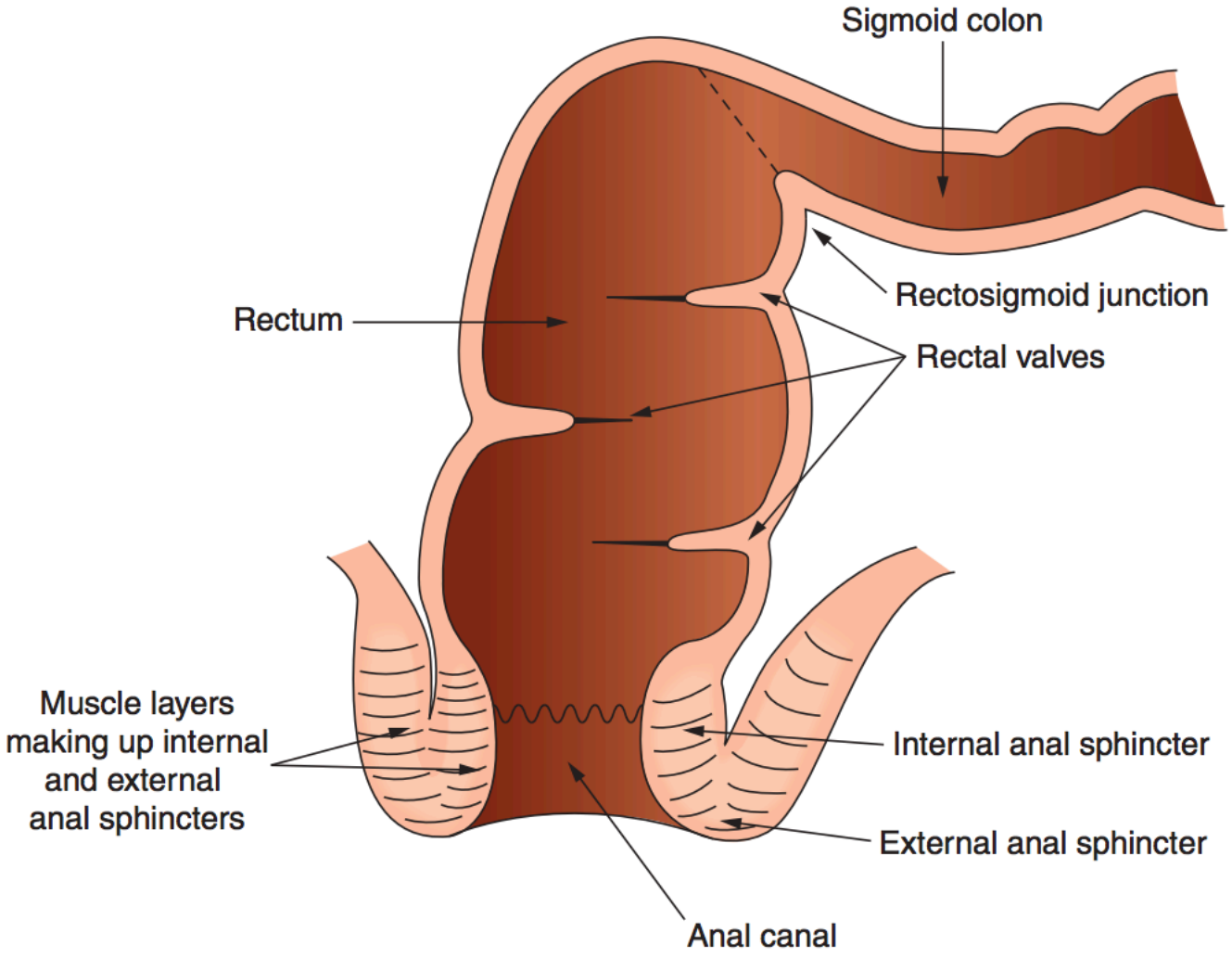
Large intestinal motility

- Cecum and proximal colon: absorption
- Distal colon: storage

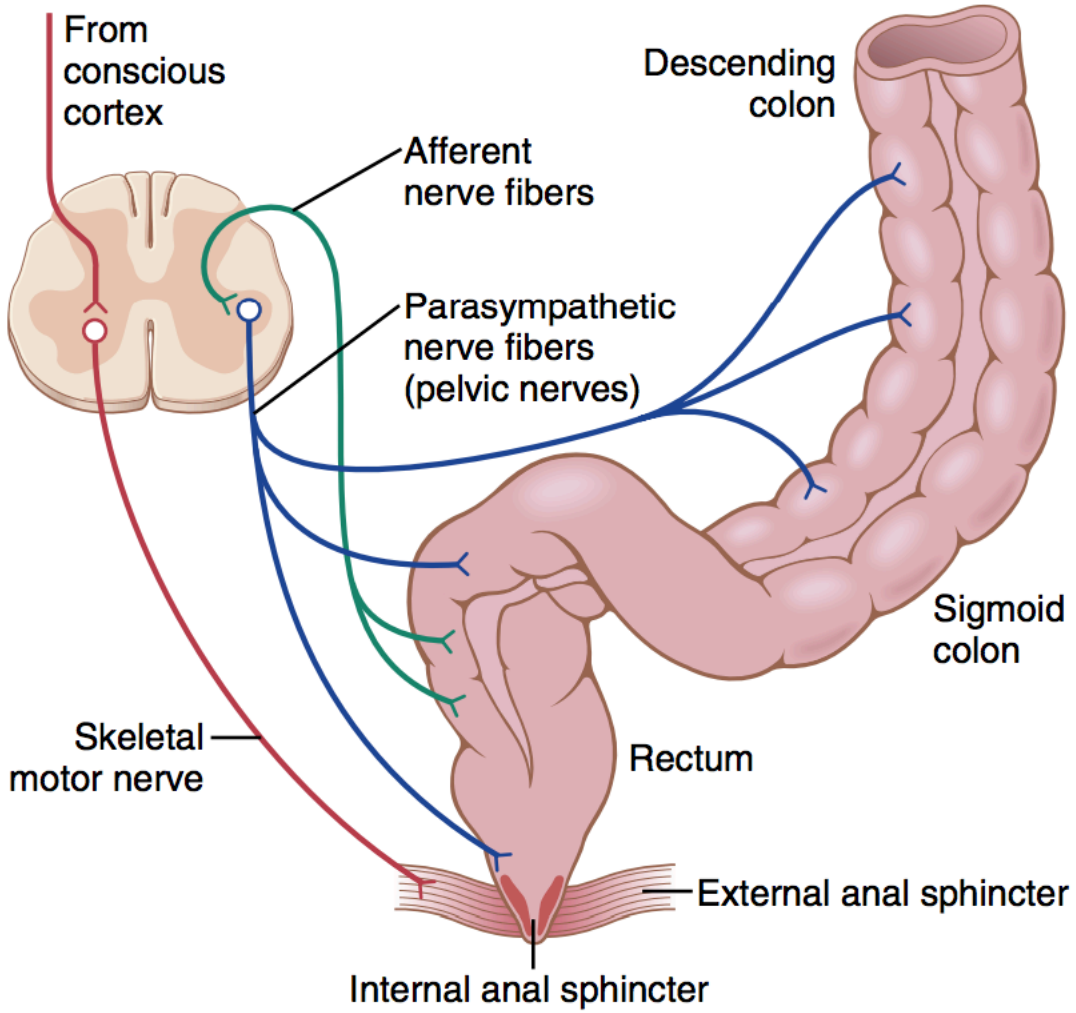
- Mass movements occur 1 to 3 times/day and cause the colonic contents to move distally for long distances (e.g., from the transverse colon to the sigmoid colon).
  - Especially for about 15 minutes during the first hour after eating breakfast.
  - Usually persists for 10 to 30 minutes
  - Facilitated by gastrocolic or duodenocolic reflex

Physiology of defecation

Anatomy and innervation of the rectum and anal canal



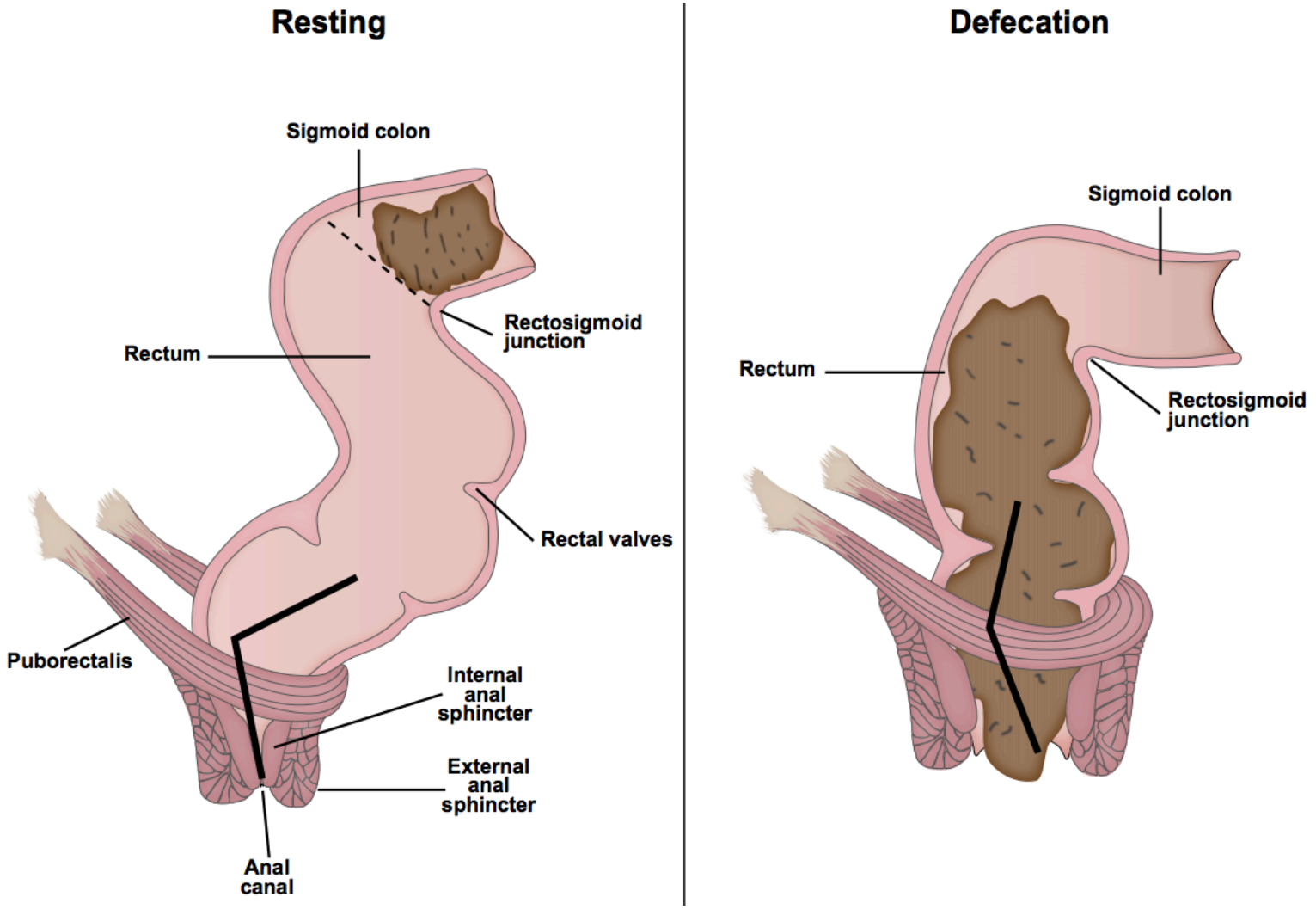
Gastrointestinal Physiology. Lange Medical Books/McGraw-Hill, 2006.



Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

Defecation

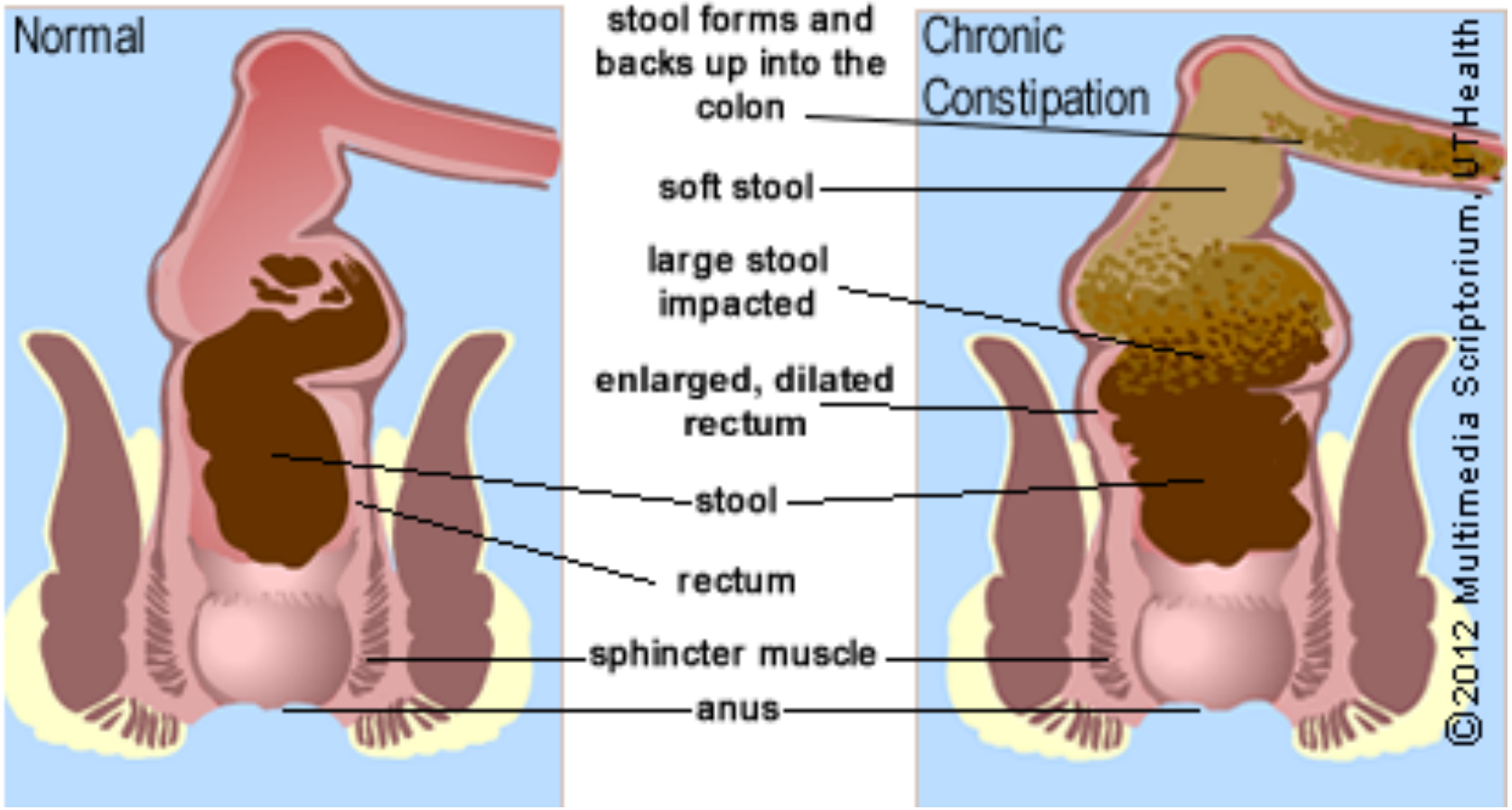
- Rectum fills with fecal material, it contracts and the internal anal sphincter relaxes
- External anal sphincter contract to prevent defecation until convenient
- At rest, the anorectal angle is approximately 90 degrees; with voluntary squeeze, the angle becomes more acute, about 70 degrees; and during defecation the angle becomes obtuse, about 110 to 130 degrees



Physiology and Pathophysiology of Digestion. Morgan & Claypool; 2018.



Clinical application: chronic constipation



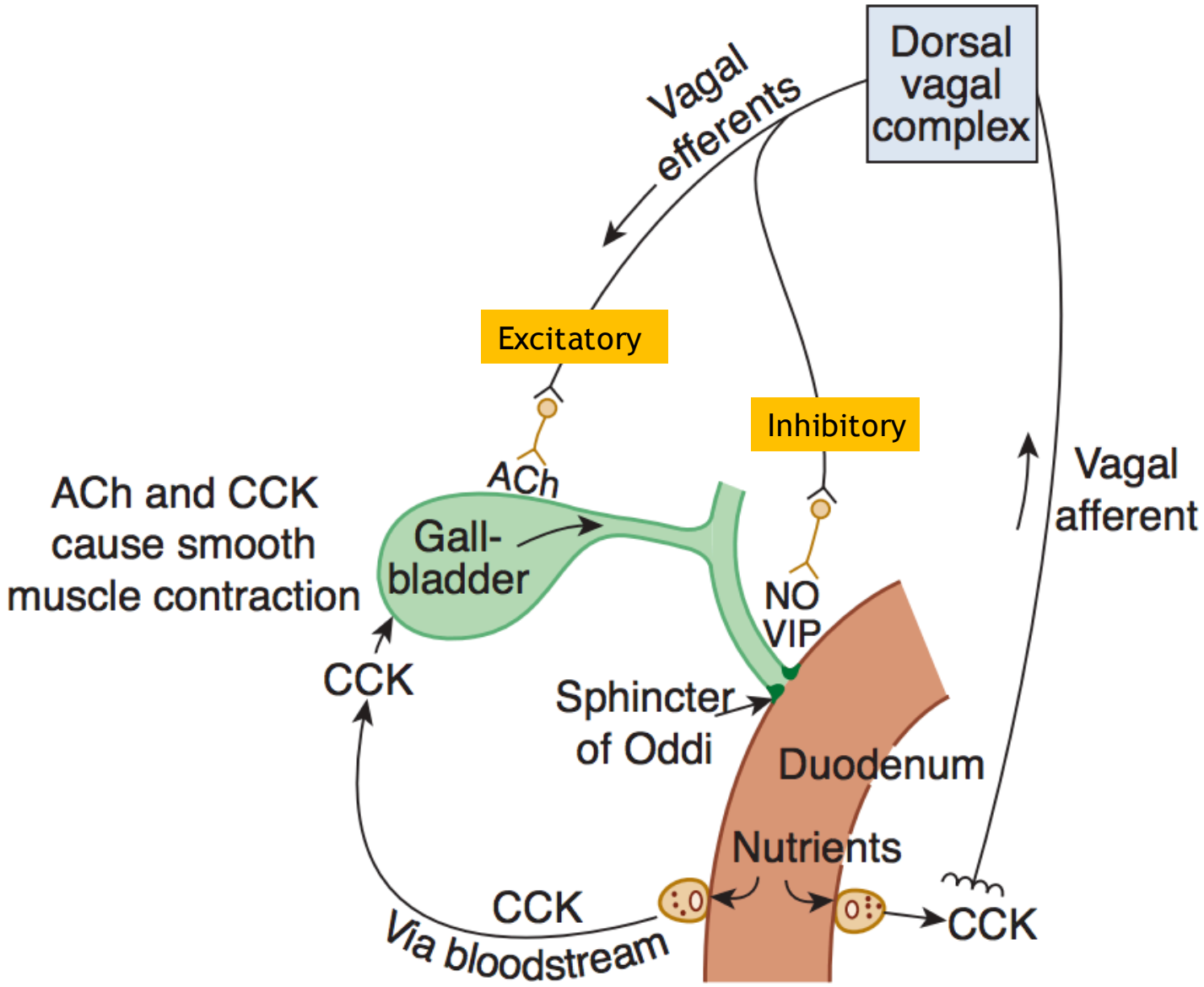
- Slow transit constipation
- Rectal evacuation disorder

Gallbladder and biliary tract motility

Gallbladder contraction & sphincter of Oddi relaxation

- Postprandial gallbladder contraction coincides with gastric emptying.
- Meal in duodenum > release of CCK
- CCK
  - 1) travels through the bloodstream, and binds directly to CCK-A receptors on gallbladder smooth muscle cells > GB contraction
  - 2) initiate a vagovagal reflex that releases ACh at gallbladder synapses > GB contraction
  - 3) initiate a vagovagal reflex that releases NO, VIP at sphincter of Oddi > relaxation

Neurohumoral control of gallbladder contraction and biliary secretion



Ganong's Review of Medical Physiology, 23rd ed. McGraw-Hill Medical, 2009.